Chapters 16 & 17: Monopolistic Competition & Oligopoly

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Outline

Today we will talk about:

- Monopolistic competition
  - short vs. long run
  - entry and exit
  - efficiency and externalities
- Advertising
- Oligopoly
  - Oligopolies and Duopolies
  - Oligopoly production choices
  - Collusion and Cartels
  - The Prisoner's Dilemma

Monopolistic Competition

What is monopolistic competition?

Many firms sell products that are similar but not identical

- Many buyers and sellers
- Products are *similar but not identical*
- Free entry and exit (in the long run)

What are some examples?

Monopolistic Competition

Just like any other type of market...

Firms maximize profit when they set $MR = MC$!

What will the firms demand curve look like?

Monopolistic Competition

Just like any other type of market...

Firms maximize profit when they set $MR = MC$!

What will the firms demand curve look like?

- Firms are not price takers - they have some market power due to differentiated products
- Firm demand will resemble that of a monopoly
- It is downward sloping!

Let's draw this market...

Short vs. Long run

In the short-run:

- If $P > ATC \implies$ positive π
- If $P < ATC \implies$ loss As long as $P > AVC$, the firm continues to operate

In the long-run:

- When firms make losses in the SR $\implies$ exit
  - $\rightarrow$ increases demand for each existing firm
  - $\rightarrow \uparrow P \quad \& \uparrow \pi$
- When firms make positive $\pi \implies$ new firms enter & produce close substitutes
  - $\rightarrow$ reduces demand for each existing firm
  - $\rightarrow \downarrow P \quad \& \downarrow \pi$

**** entry/exit $\implies$ all firms earn 0 economic π in LR!! ****

In the long run...

Two important characteristics of monopolistic competition:

1. As in a monopoly market: $MC < P$ - **Markup**
   - maximizing π means setting $MR = MC$, and $MR <$ demand curve, which gives us price
   - So $MR < P$
2. As in a competitive market: $P = ATC$ - **No profits**
   - free entry and exit drive economic profit to 0

The optimal Q for a monopolistically competitive firm is Q such that:

1. $MR = MC$
2. ATC is tangent to (kisses) the firms demand curve

Monopolistic vs. Perfect Competition

Two key differences:

Excess capacity:

The **efficient scale** is the Q which minimizes ATC

- A monopolistically competitive firm has excess capacity:
  - They could $\uparrow Q$ and that would $\downarrow ATC$

Markup:

A monopolistically competitive firm operates where:

- ATC is decreasing $\rightarrow$ where $MC < ATC$
- $\implies MC < P$
- firms want to induce additional demand

Monopolistic competition and welfare

Externality: uncompensated impact of one person/firms actions on a bystander person/firm

- Positive externality: surplus of bystander goes up
- Negative externality: surplus of bystander goes down

How does firm entry affect a monopolistically competitive market?

- **Product-variety externality:** positive externality on consumers
- **Business-stealing externality:** negative externality on sellers

Advertising

What is the point of advertising?

- *Informs*: conveys prices and locations of goods, introduces new products
- *Misleads*: convinces that substitutes are more distinct than they really are
- *Signals*: willingness to spend money to advertise indicates high quality

Is this good or bad for welfare? We don't really know!

What is an Oligopoly?

An **oligopoly** is a market in which:

- there are only a few sellers
- they offer similar or identical products

A **Duopoly** is an oligopoly with only 2 sellers

- We usually consider duopolies since they have the same outcomes/properties as larger oligopolies

What are some examples?

Our different types of markets

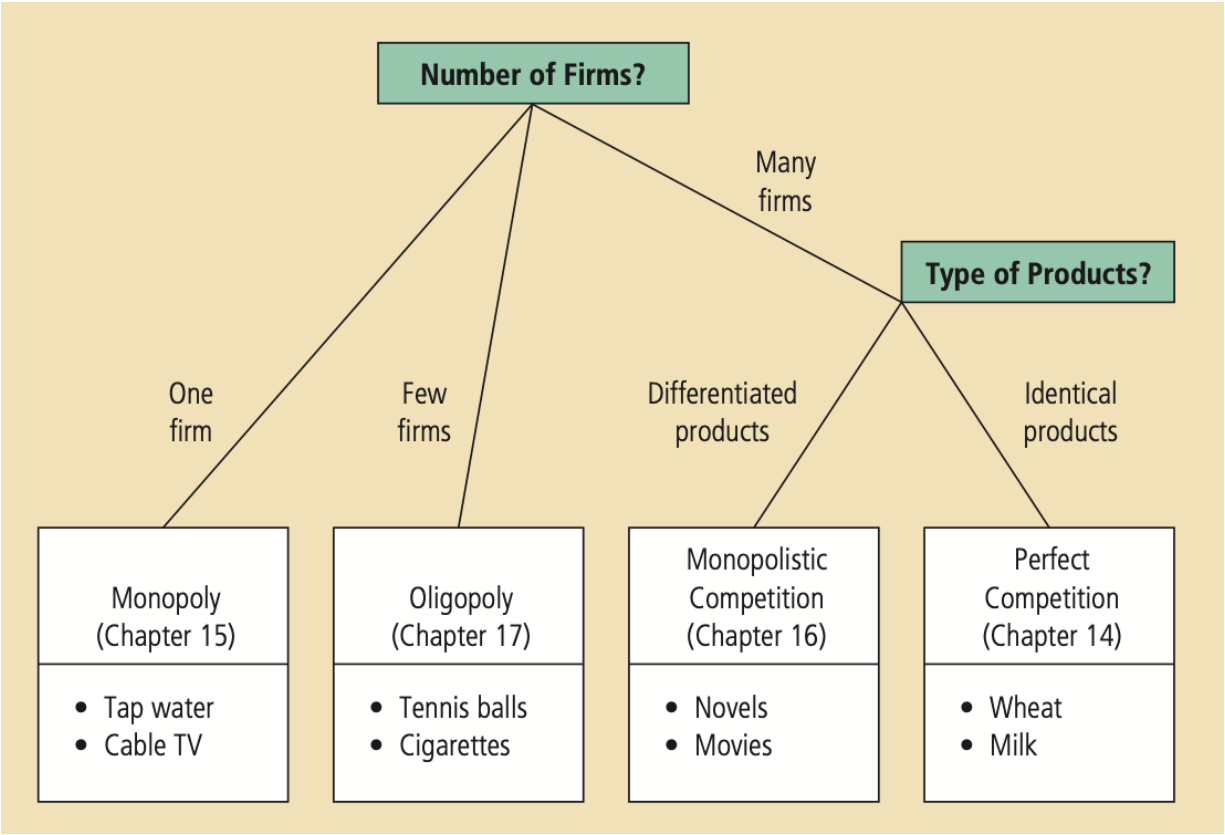

Figure 1: Caption

Oligopoly profits

Verizon and Xfinity are a duopoly for internet service in Baltimore. Let's assume *MC* of providing service to each customer is \$40. The demand schedule for internet service in Baltimore is:

Price	Quantity
\$180	5,000 customers
\$160	6,000
\$140	7,000
\$120	8,000
\$100	9,000
\$80	10,000
\$60	11,000
\$40	12,000

What would the market *P* and *Q* be if this was a *monopoly market*? What about a *perfectly competitive market*?

How does an Oligopoly determine P^* & Q^*?

When there are a small number of firms, as in an oligopoly, those firms can collude...

Collusion is when:

oligopoly firms make an agreement over prices and production

A **cartel** is:

A group of oligopoly firms which are colluding together

Given these definitions,

- Can Verizon and Xfinity collude?
- Would they want to? Why?
- What would this look like?

We know that the monopoly P & Q result in the highest possible profit for the monopoly firm...

- Verizon and Xfinity can act together as a single monopoly by colluding!
- How? $\rightarrow$ Set the sum of their production as the monopoly quantity and charge the monopoly price
- Choose how to split production among the two firms

Let's say Verizon and Xfinity collude and produce the monopoly P & Q and split production 50-50:

Price	Quantity	Market π	Verizon π	Xfinity π
\$180	5,000	\$700,000	\$350,000	\$350,000
\$160	6,000	\$720,000	\$360,000	\$360,000
\$140	7,000	\$700,000	\$350,000	\$350,000
\$120	8,000	\$640,000	\$320,000	\$320,000
\$100	9,000	\$540,000	\$270,000	\$270,000
\$80	10,000	\$400,000	\$200,000	\$200,000
\$60	11,000	\$220,000	\$110,000	\$110,000
\$40	12,000	\$0	\$0	\$0

If Verizon decide's to increase the quantity of internet service they provide by 1,000, but Xfinity makes no change, the new market production is 7,000:

Price	Quantity	Market π	Verizon π	Xfinity π
\$180	5,000	\$700,000	\$350,000	\$350,000
\$160	6,000	\$720,000	\$360,000	\$360,000
\$140	7,000	\$700,000	\$400,000	\$300,000
\$120	8,000	\$640,000	\$320,000	\$320,000
\$100	9,000	\$540,000	\$270,000	\$270,000
\$80	10,000	\$400,000	\$200,000	\$200,000
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Verizon broke the agreement by increasing production, and is now earning more profit than under the collusion agreement. Xfinity is left worse off.

But if Xfinity knows Verizon will increase production by 1,000, they will want to increase their production by the same amount to mitigate the loss of profit:

Price	Quantity	Market π	Verizon π	Xfinity π
\$180	5,000	\$700,000	\$350,000	\$350,000
\$160	6,000	\$720,000	\$360,000	\$360,000
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But Verizon still has incentive to increase their production without telling Xfinity, and so Xfinity will also want to increase production...

Nash Equilibrium

The interaction between Verizon and Xfinity tells us something about the strategic behavior of oligopolies...

A nash equilibrium is:

A set of strategies such that economic actors interacting together each choose their best strategy given the strategies that the others have chosen

When Verizon and Xfinity act in their own self interest by increasing production, they lose out on profit.
$\rightarrow$ So they would rather collude to produce monopoly P & Q

But will they cooperate? Is this easy?

The Prisoner's Dilemma

Let's consider a "Game" called the Prisoner's Dilemma:

- 2 criminals (Bonnie and Clyde) stole $100 and were caught
- Each has the option: Confess or Stay silent
- There are 3 possible outcomes
  - Both stay silent = they keep $100 and split it 50-50
  - One confesses and the other remains silent = silent one gets nothing while confessor gets $60
  - Both confess: they split the $60 reward 50-50

What does this look like? How do we find the nash equilibrium?

Cooperation

- Cooperation is hard to maintain! Even when it would make you better off
- The same principles apply to oligopolies and their choice of production
  - collusion at the monopoly P & Q is rational for both providers
  - BUT each firm has an incentive to cheat
  - and each firm anticipates the other cheating

So do oligopoly firms ever actually form cartels?

Externaltities

Recall... **Externality:** uncompensated impact of one person/firms actions on a bystander person/firm

- Positive externality: surplus of bystander goes up
- Negative externality: surplus of bystander goes down

We can **internalize an externality**

$\rightarrow$ alter incentives so that people take the external impacts of their actions into account

- Externalities are often cited as reasons for government intervention:
  - **Corrective/Pigouvian tax:** a tax designed to alter buyer demand through changes in price to bring the market equilibrium quantity closer to the social optimal level
  - Forces individuals to internalize the external impacts of their actions
  - Examples?

Internalizing an externality

Consider the market for cigarettes

- Second hand smoke is bad for bystanders $\rightarrow$ cigarettes cause a negative externality
- So the socially optimal quantity $Q^{SO} < Q^*$, the equilibrium quantity

The government can tax cigarettes to reduce the equilibrium quantity to the socially optimal level

** What does this look like?